

## B The user manual for the Client scanner

The Client scanner is the console application, that is distributed as the single executable file named `PROXY.JAR` together with the folder named `LIB`, which contains the necessary third-party libraries.

### B.1 The application requirements

The Client scanner has following requirements:

- The Java Runtime Environment 1.8.
- Java Cryptography Extension (JCE) Unlimited Strength Jurisdiction Policy Files 8
- The Linux or Windows operating system.

### B.2 Before the first run

Client scanner requires extension for JRE 1.8 named *Java Cryptography Extension (JCE) Unlimited Strength Jurisdiction Policy Files 8*. Mentioned extension can be downloaded from <http://www.oracle.com/technetwork/java/javase/downloads/jce8-download-2133166.html> as an archive named `JCE_POLICY-8.ZIP`. Inside this archive you will find a file `README.TXT`, that contains installation instructions.

### B.3 The first run

To run the Client scanner, the file `PROXY.JAR` and the folder `LIB` must be in the same location. The correct file structure may look like this:

- `/folder/lib/`
- `/folder/proxy.jar`

Now, you can run the file `PROXY.JAR` with the Java Runtime Environment. The Client scanner will automatically access the `LIB` folder on the expected location.

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On the Linux operating system, the Client scanner can be launched via the following command:

```
java -jar proxy.jar
```

When you run the Client scanner for the first time, it will immediately exit and create the XML file named (CONFIGURATION.XML). You will find the mentioned file in the folder, where PROXY.JAR is located.

### B.4 The regular run

When you already ran the Client scanner for the first time and you prepared multiple XML configuration files, you may run the Client scanner with the optional argument, that specifies the name of the configuration file:

```
#will use specified production.xml file  
java -jar proxy.jar production.xml
```

```
#will use default configuration.xml file  
java -jar proxy.jar
```

### B.5 The configuration

The configuration consists of the single XML file. There, you will find the five main setting's categories – *directives*, *protocols*, *cipherSuites* and *certificates*.

#### B.5.1 The section *directives*

In the category *directives*, you must specify following basic values:

- **debug** – Should the Client scanner print the debug messages or not?
- **localPort** – The local port, where will be the Client scanner listening to the incoming communication.
- **remoteHost** – The IP address or DNS name of the remote host.
- **remotePort** – The remote port for communication with the remote host.

### **B.5.2 The section *protocols***

In the category *protocols*, you are telling, which protocol can, must be or must not be supported. The available protocols (in the order from the oldest to the newest) are *SSLv2*, *SSLv3*, *TLS 1.0*, *TLS 1.1* or *TLS 1.2*. In the case of *SSLv2*, only *SSLv2* handshake is being checked, full proxy communication over *SSLv2* is not supported due to limitation of the JRE 1.8.

### **B.5.3 The section *cipherSuites***

The category *cipherSuites* defines, which the cipher suites can, must be or must not be offered during the *SSL/TLS* handshake. To correctly recognize the cipher suite, you must state its name. There are currently the several naming conventions, so to minimize a risk of the user's mistake, the supported ciphers suites are the internal part of the Client scanner. The mentioned list of currently supported cipher suites is available directly in the source code as the content of `/SRC/MAIN/RESOURCES/CIPHERS.XML` file. This list can be expanded by adding a new record with the proper hexadecimal value (as defined in the RFCs) and the name in your selected naming convention.

### **B.5.4 The section *certificates***

The category *cipherSuites* defines certificates, which must be or must not be used to secure communication. The proxy server expects each certificate as the Java KeyStore file, which actually allows you to specify whole certificate chain. At least one certificate with mode *mustBe* is required, because that certificate will be offered to the client during some tests. Each Java KeyStore file has to contain exactly one private key.

## **B.6 Running the application**

### **B.6.1 The prerequisites**

The Client scanner is primarily intended for the software developers. To use it correctly, you must be able to do the small modification in the application, that will be tested. The tested application must be

repeatedly trying to establish the connection to the remote server, even if the previous attempt already was or wasn't successful. There are no precise instructions how to modify your application, because the implementation of the required modification depends on many factors.

### B.6.2 Launching the application

When the mentioned XML file contains the required values, the Client scanner is simple to use. Just run the application. It will automatically parse the XML file, launch listening on the defined port and analyse the SSL/TLS traffic coming through the proxy. After completion of all designated tests, the Client scanner will automatically invoke its shutdown and the report with test's results is created.

### B.6.3 Example

Let's imagine, you want to test your web browser. In the file `CONFIGURATION.XML` you may specify *localPort* as **2222**, *remoteHost* as **www.muni.cz** and *remotePort* as **443**. Now, when you run the application, you must in your web browser manually and repeatedly load URL `https://localhost:2222`. You may see correctly loaded homepage of the website **www.muni.cz** or some error message presented by your browser (it is worth to mention, that both states are expected behavior). After last automated test, you may close your web browser and examine report of the test's results.

## B.7 The report of results

After each quit of the Client scanner, a new HTML file is created in the same directory, where `PROXY.JAR` is contained. Such the file contains details about captured the SSL/TLS traffic. Please, be aware, that the report in the detail primary talks about the discovered vulnerabilities. The secure results are described briefly or not mentioned at all.